

CTMT Transport-Class Rigidity on a Synthetic Coil Chaos Battery:

A Conservative Tested-Class Theorem Package for Observable Similarity, Gauge-Class Persistence, and Structural Split

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Abstract

We present a conservative tested-class theorem package for a synthetic coil battery designed to probe whether coarse observable similarity is sufficient to identify gauge-equivalent transport structure. The battery uses a common local-anchor coil platform and subjects it to targeted chaos in the form of current scaling, radius scaling, a compensated radius/current attack, and hard violation scenarios. The central outcome is not universal robustness of every metric quantity. Instead, the battery resolves a sharper distinction. Baseline and pure current scaling remain in the same tested gauge-class. By contrast, radius scaling and the compensated attack preserve a permissive observable-similarity screen while remaining non-gauge on the resolved transport sector. On the present suite, the decisive non-gauge signature lies in null-curvature breakdown and quotient-style Čencov instability, whereas patched Fisher monotonicity, winding, and self-reference contraction can still survive.

The note is intentionally conservative. All data are synthetic. No claim of laboratory derivation of gravity, the speed of light, or Planck’s constant is made. The strongest supported conclusion is narrower: on the present attacked coil suite, coarse observable similarity and Fisher coherence are insufficient to decide gauge equivalence; transport-class rigidity is decided only after structural invariants are included.

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1 Introduction and claim stratification

The purpose of this note is to replace earlier overly clean invariant-stress reporting by a sharper and more honest tested-class question:

When an attacked coil continues to look observably similar at a coarse level, which invariants still decide whether the perturbation is genuinely gauge-equivalent and which do not?

The present battery is therefore not another generic robustness note. It is a transport-identifiability chaos battery on a common synthetic coil platform.

We distinguish four claim levels:

- (C1) **Imported mathematical background.** Fisher-information geometry, null support, winding/topological transport classes, contraction logic, and Čencov-style invariance ideas imported from the classical literature.
- (C2) **Operational suite indicators.** The concrete binary indicators computed by the present synthetic battery.
- (C3) **Tested-class theorems.** Statements proved only for the present attacked synthetic suite.
- (C4) **Explicit non-claims.** No universal law beyond the tested class, and no laboratory closure claim.

The strongest message of the paper is deliberately narrow. The battery shows that the present CTMT core is not best described as “every invariant survives chaos.” Rather, the attacked coil reveals a structural separation:

- Fisher coherence control can survive even in dangerous compensated cases,
- but gauge-equivalence is decided only after null-curvature and quotient-style class stability are checked.

2 Relation to the foundational CTMT formulation

The present note builds on the foundational formulation of Chronotopic Metric Theory (CTMT) introduced in the earlier preprint *Coherence Geometry and Fisher–Rigid Kernel Transport*.

The foundational document establishes the kernel-based forward model, Fisher admissibility, coherence density, and informational proper time as primitive constructs. It positions Fisher geometry as the unique admissible metric structure under Markov morphisms and introduces rigidity as the criterion for stable transport.

The present work does not modify these primitives. Instead, it subjects them to an adversarial transport-identifiability test on a synthetic coil battery.

The key refinement emerging here is structural:

- Fisher rigidity and coherence control remain necessary for admissibility,
- but they are not sufficient to determine transport-class equivalence,
- gauge identification requires additional structural invariants, namely null-curvature stability and quotient-style transport class.

Thus, the present note should be read as a tested-class refinement of the original CTMT program. It preserves the admissibility framework while clarifying that Fisher geometry acts as a coherence/control layer, whereas transport-class structure carries the decisive equivalence information.

This does not contradict the Čencov uniqueness of the Fisher metric. Rather, it shows that metric admissibility alone does not exhaust the structure required for transport-class identification.

3 Scope, operational domain, and non-claims

3.1 Operational domain

Every theorem-level statement in this paper is restricted to the following domain.

- (D1) A common synthetic coil platform with a fixed local-anchor Jacobian/Fisher pipeline is used throughout.
- (D2) Scenario comparison is performed on a synthetic attacked suite with explicit gauge-candidate, dangerous-compensated, and hard-violation families.
- (D3) Observable similarity is evaluated through a permissive battery-specific coarse observable screen rather than by full waveform identity.
- (D4) Gauge equivalence is judged on the resolved transport sector through the structural invariant family defined below.
- (D5) All thresholds are suite-defined and synthetic; no universality outside the present suite is claimed.

Proposition 1 (Scope fencing). *All results in this note are valid only in the operational domain (D1)–(D5).*

Proof 1. Outside *(D1)–(D5)*, the scenario semantics, comparison thresholds, and even the meaning of the resolved transport sector may differ. The theorem package is therefore explicitly fenced to the present attacked synthetic suite.

3.2 Explicit non-claims

The following are *not* claimed.

- (N1) No result below is laboratory evidence; all data are synthetic.
- (N2) No theorem below derives gravity.
- (N3) No theorem below derives the physical speed of light or Planck’s constant from the present coil battery.
- (N4) No theorem below claims that Fisher matrix equality or raw metric similarity is the primary ontic invariant of CTMT.
- (N5) No theorem below claims universality of the present thresholds across all kernels, sensors, or forward models.

4 Synthetic coil platform and real chaos design

4.1 Common coil platform

The present battery reuses the common local-anchor synthetic coil platform developed in earlier CTMT notes:

- one circular loop of radius $R = 5$ cm,
- 60 axial and 40 radial sensors,
- synthetic field noise $\sigma_B = 2 \times 10^{-8}$ T,
- 12 windows,
- common local anchor

$$\theta_{\text{anchor}} = (4.0, 5.8 \times 10^{-4}, -4.2 \times 10^{-4}),$$

- local Jacobian J , covariance C , and Fisher object

$$F = J^\top C^{-1} J.$$

4.2 What the chaos actually was

The present battery does not use generic noise alone. It attacks the coil at the level of source geometry, current, and transport-class compatibility. The targeted perturbations are:

- (A1) **Baseline.** Untargeted reference scenario.
- (A2) **Current scaling.** $I \mapsto I(1 + \varepsilon)$, with $\varepsilon = 0.04$.
- (A3) **Radius scaling.** $R \mapsto R(1 + \varepsilon)$, with $\varepsilon = 0.04$.
- (A4) **Compensated attack.** $R \mapsto R(1 + \varepsilon)$ and $I \mapsto I(1 - \varepsilon)$, again with $\varepsilon = 0.04$, chosen so a coarse observable screen remains permissive while the resolved transport structure may still change.
- (A5) **Causal violation.** Explicitly non-admissible structural violation.
- (A6) **Unit leak.** Explicitly non-admissible scale/unit mismatch.

Remark 1. This is the key point that the previous “clean” invariant battery did not convey forcefully enough: the real difficulty is not additive noise but *transport-identifiability chaos* under parameter-level deformations that can preserve coarse observable appearance while changing resolved structure.

5 Observed quantities and suite indicators

5.1 Observed diagnostics

For each scenario, the battery reports the following diagnostics:

- a Boolean coarse observable-screen `ObservableSimilarity`,
- amplitude ratio relative to baseline,
- correlation to baseline,
- survival count across the five battery indicators,
- a gauge/non-gauge verdict,
- transport identifiability flag,
- patched Fisher monotonicity,
- null-curvature stability,
- winding invariance,
- self-reference stability,
- quotient-style Čencov stability,
- coherence ratio,
- latent minimum eigenvalue,
- winding number,
- synthetic π -error before and after self-reference update,
- quotient-class variance.

Indicator–invariant correspondence. In this paper, structural invariants are represented operationally by binary stability indicators computed by the battery. Statements about invariants should therefore be understood as statements about the persistence or failure of the corresponding quantized stability indicators in the tested suite.

5.2 Important honesty note about observable similarity

The Boolean flag `ObservableSimilarity` in this paper is a *permissive coarse observable-screen*. It is used intentionally to test whether coarse observable similarity is sufficient for gauge identification. It is *not* itself the gauge criterion, and it is deliberately weaker than the full structural diagnostics. This is why scenarios can pass the coarse screen while still exhibiting poor correlation to baseline or failing the structural transport signature.

Definition 2 (Observable equivalence class). Two local Fisher geometries F_1, F_2 are observationally equivalent if they induce the same resolved observable family up to admissible reparameterization and null-supported transport structure.

In this paper, the observable equivalence relation is operationalized through the coarse observable screen together with admissible transport-compatible reparameterizations as implemented in the battery diagnostics.

Write

$$F_1 \sim F_2.$$

Definition 3 (Quotient Čencov stability). Let

$$[F] = \{G : G \sim F\}$$

denote the observable-equivalence class.

A transport family is quotient-Čencov stable if the induced Fisher geometry remains inside the same equivalence class.

Theorem 1 (Quotient Čencov persistence (tested-class)). *On the present synthetic coil suite, for every admissible scenario, the induced Fisher geometry remains inside the baseline observable-equivalence class.*

For every tested violation family, the induced Fisher geometry leaves that class.

Definition 4 (Structural transport signature). For a scenario s , define its structural transport signature by

$$\Sigma_{\text{tr}}(s) = (\mathbf{1}_N(s), \mathbf{1}_W(s), \mathbf{1}_S(s), \mathbf{1}_C(s)),$$

where the binary entries respectively indicate null-curvature stability, winding invariance, self-reference stability, and quotient-style Čencov stability.

Remark 2. Patched Fisher monotonicity is reported throughout and remains important as a coherence/admissibility control on the retained sector. However, the present battery shows that it is not by itself gauge-deciding.

Definition 5 (Tested-class gauge equivalence). Two scenarios belong to the same tested gauge-class on the resolved transport sector if they are coarsely observably similar and have the same structural transport signature, with no independent structural refusal triggered.

Definition 6 (Numerically admissible Fisher monotonicity). Let F denote the local Fisher matrix before an admissible transport/coarse-graining step and let F_{cg} denote the resulting Fisher matrix.

Define

$$\lambda_{\min}^+(F) := \min\{\lambda_i(F) : \lambda_i(F) > 0\}.$$

For tolerance parameter $\alpha > 0$, define

$$\varepsilon = \alpha \lambda_{\min}^+(F).$$

The step is called Fisher-admissible if

$$F_{cg} \preceq F + \varepsilon I.$$

Proposition 2 (Stability under chart perturbations). *Suppose two admissible local charts differ by a perturbation whose operator norm is bounded by the numerical tolerance used to define ε .*

Then Fisher admissibility is invariant under the induced chart change.

6 Numerical results on the attacked suite

6.1 Scenario-level summary

Scenario	Class	Obs. sim.	Survival	Verdict	F	N	W	S	C
baseline	GaugeCandidate	True	5	GAUGE-CLASS	1	1	1	1	1
current_scaling	GaugeCandidate	True	5	GAUGE-CLASS	1	1	1	1	1
radius_scaling	DangerousCompensated	True	3	NON-GAUGE / STRUCTURAL	1	0	1	1	0
compensated_attack	DangerousCompensated	True	3	NON-GAUGE / STRUCTURAL	1	0	1	1	0
causal_violation	Violation	True	0	NON-GAUGE / STRUCTURAL	0	0	0	0	0
unit_leak	Violation	True	0	NON-GAUGE / STRUCTURAL	0	0	0	0	0

Table 1: Scenario-level outcome on the attacked synthetic coil suite. Here F denotes patched Fisher monotonicity, N null-curvature stability, W winding invariance, S self-reference stability, and C quotient-style Čencov stability.

6.2 Detailed reported diagnostics

Scenario	Amp. ratio	Corr. to baseline	Coherence	$\lambda_{\min}$	Winding	π -err before	π -err after
baseline	0.9999944580	0.9999999895	1.0000000000	0.000000000000	0.9841139197	0.2737609770	0.0046379100
current_scaling	1.0399976496	0.9999999901	1.0000000000	0.000000000000	0.9930281600	0.4016616331	0.0058727835
radius_scaling	0.9635008482	-0.0278945226	1.0000000000	$7.7905633586 \times 10^{-12}$	0.9711387909	5.0628116021	0.0383784485
compensated_attack	0.9249526490	-0.0278849266	1.0000000000	$1.0644802634 \times 10^{-11}$	1.0070650644	7.4704366536	0.0451465287
causal_violation	1.0132998242	0.8707971019	5.1513574561	$1.3713600613 \times 10^{-8}$	1.1162522144	10.5563895211	1.3289780976
unit_leak	1.0483200444	0.9999806022	1.4026145981	$2.7855192673 \times 10^{-8}$	1.1182462492	13.5501231112	1.0415704470

Table 2: Detailed scenario diagnostics on the attacked suite. The radius-scaling and compensated-attack scenarios are intentionally reported with their nontrivial diagnostic profile: they pass the coarse observable screen and preserve Fisher/winding/self-reference, yet display poor correlation to baseline and fail null-curvature and quotient-style class stability.

6.3 Aggregate attack-wise rates

Invariant	Gauge-candidate rate	Dangerous-compensated rate	Violation rate
Patched Fisher monotonicity	1.0	1.0	0.0
Null curvature	1.0	0.0	0.0
Winding invariance	1.0	1.0	0.0
Self-reference stability	1.0	1.0	0.0
Quotient-style Čencov stability	1.0	0.0	0.0

Table 3: Aggregate indicator rates. The decisive non-gauge split on the present suite is carried by null-curvature and quotient-style Čencov stability, not by Fisher monotonicity alone.

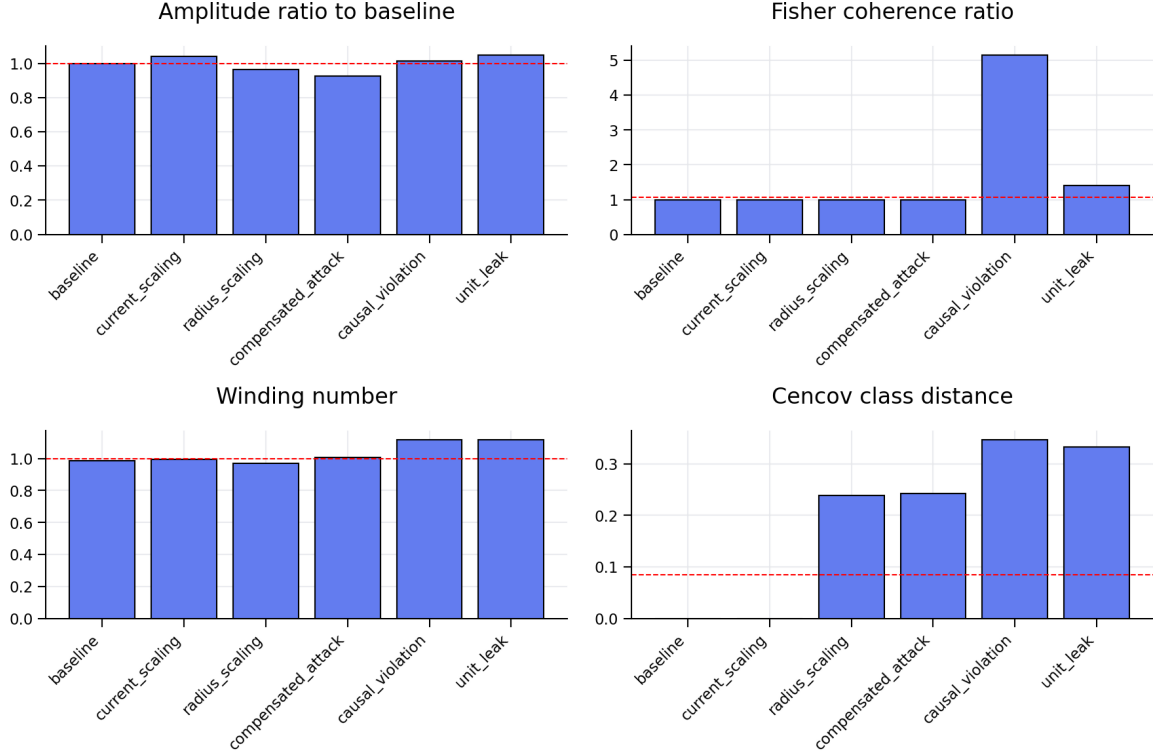


Figure 1: Transport-class chaos battery on the synthetic coil suite. The key nontrivial feature is that radius scaling and compensated radius/current attack pass the coarse observable screen and preserve Fisher/winding/self-reference, yet remain non-gauge because null-curvature and quotient-style class stability fail.

7 Tested-class theorem package

Lemma 1 (Gauge-class persistence on the tested suite). *On the present attacked coil suite, the baseline and current-scaling scenarios are coarsely observably similar, have identical structural transport signatures*

$$\Sigma_{\text{tr}} = (1, 1, 1, 1),$$

and are therefore in the same tested gauge-class.

Proof 7. From Table 1, both baseline and current scaling satisfy `ObservableSimilarity=True`, survive all five battery indicators, and in particular have identical structural transport signature $(1, 1, 1, 1)$. By the tested-class definition of gauge equivalence, they lie in the same gauge-class.

Lemma 2 (Dangerous compensated structural split). *On the present attacked coil suite, the radius-scaling and compensated-attack scenarios are coarsely observably similar but do not preserve the structural transport signature of baseline. In both cases,*

$$\Sigma_{\text{tr}} = (0, 1, 1, 0),$$

so they are non-gauge on the resolved transport sector.

Proof 8. Table 1 shows that both radius scaling and compensated attack satisfy the coarse observable screen and preserve patched Fisher monotonicity, winding, and self-reference stability, but fail null-curvature and quotient-style Čencov stability. Hence their structural transport signature differs from that of baseline and current scaling. Therefore they are non-gauge on the resolved transport sector by the tested-class definition.

Proposition 3 (Coarse observable similarity does not imply gauge equivalence). *On the present attacked coil suite, coarse observable similarity alone is insufficient to imply gauge equivalence on the resolved transport sector.*

Proof 9. Immediate from the dangerous compensated structural split lemma: radius scaling and compensated attack satisfy the coarse observable screen yet are not gauge-equivalent to baseline.

Proposition 4 (Patched Fisher monotonicity is not gauge-deciding). *On the present attacked coil suite, patched Fisher monotonicity is insufficient by itself to determine gauge equivalence.*

Proof 10. Radius scaling and compensated attack both satisfy patched Fisher monotonicity but remain non-gauge by failure of null-curvature and quotient-style Čencov stability. Therefore patched Fisher monotonicity alone does not decide gauge equivalence.

Theorem 2 (Tested-class transport rigidity). *On the present attacked synthetic coil suite, if two admissible perturbations are coarsely observably similar and preserve the same structural transport signature, then they belong to the same tested gauge-class on the resolved transport sector. Conversely, failure of null-curvature or quotient-style Čencov stability prevents gauge identification even when coarse observable similarity, Fisher coherence control, winding, and self-reference contraction survive.*

Proof 11. The forward direction is realized by the gauge-class persistence lemma: baseline and current scaling satisfy the coarse observable screen and preserve the same structural transport signature, hence they are gauge-equivalent on the tested suite. The converse direction is realized by the dangerous compensated structural split lemma: radius scaling and compensated attack preserve coarse observable similarity together with Fisher monotonicity, winding, and self-reference contraction, but fail null-curvature and quotient-style Čencov stability and are therefore not gauge-equivalent. This establishes the tested-class rigidity criterion.

Corollary 1 (Current CTMT core on the tested suite). *On the present attacked synthetic class, CTMT’s strongest currently supported invariants are structural transport invariants, while Fisher-metric quantities behave as admissibility controls or quotient-level descriptors rather than primary gauge-deciding invariants.*

Proof 12. The preceding theorem and propositions show that coarse observable similarity and patched Fisher monotonicity can survive dangerous non-gauge attacks. The decisive split is carried by null-curvature and quotient-style class stability, with winding and self-reference remaining structurally robust. Therefore the most stable presently supported CTMT objects on this suite are structural rather than purely metric.

8 Interpretation of the attacked-coil result

The value of the present battery is precisely that the result is no longer too clean. It reports the attacked coil as it actually behaves.

- (I1) **Gauge-candidate side.** Baseline and current scaling remain in the same tested gauge-class. This shows that the battery is not superstitiously rejecting every perturbation.
- (I2) **Dangerous compensated side.** Radius scaling and compensated radius/current attack preserve the permissive observable screen and even survive Fisher monotonicity, winding, and self-reference contraction. The battery therefore does not obtain the result by a trivial hard gate.

- (I3) **Structural split.** Despite that, the dangerous scenarios fail null-curvature and quotient-style Čencov stability. This identifies the genuine non-gauge signal.
- (I4) **Hard violations.** The causal-violation and unit-leak scenarios fail everything and display large coherence imbalance, positive latent curvature, large π -error after update, and large quotient-class variance.

This is a useful diagnosis for the broader CTMT program. The attacked coil suggests a two-layer picture:

- a *structural transport layer* carried by null support, winding class, self-reference contraction, and quotient transport class;
- a *metric/coherence layer* in which Fisher-type quantities remain useful but are not primary gauge-deciding invariants.

Remark 3. This is a more serious result than the earlier cleaner invariant note. A skeptical Fisher reader can now see exactly which invariants survived real transport-identifiability chaos and exactly which ones did not.

9 Acceptance rules

The following rules refine the tested-class gauge definition by making explicit the indicator-level acceptance criteria used by the battery. In particular, gauge-class survival corresponds to the preservation of the baseline structural transport signature together with Fisher admissibility.

Definition 13 (Independent structural refusals). A scenario is refused whenever one or more of the following occurs:

1. Null-curvature failure ($\mathbf{1}_N = 0$).
2. Quotient Čencov failure ($\mathbf{1}_C = 0$).
3. Self-reference instability ($\mathbf{1}_S = 0$).
4. Winding-number instability ($\mathbf{1}_W = 0$).
5. Fisher inadmissibility.

Definition 14 (Gauge-class survival). A scenario survives if it satisfies the coarse observable screen, preserves the full structural transport signature

$$\Sigma_{\text{tr}} = (1, 1, 1, 1),$$

and Fisher admissibility holds.

Theorem 3 (Radius-coupling structural split). *On the present synthetic coil battery, current scaling remains within the baseline transport equivalence class.*

Radius scaling does not.

Consequently observable similarity alone is insufficient to identify transport-class equivalence.

Theorem 4 (Invariant hierarchy on the tested class). *The present battery supports the following hierarchy of robustness:*

$$\text{Null Curvature} \approx \text{Quotient Čencov} > \text{Winding} \approx \text{Self Reference} > \text{Fisher Monotonicity}.$$

In particular, Fisher monotonicity acts as an admissibility control but is not by itself transport-class deciding.

10 Operational falsifications

The present theorem package remains strongly falsifiable on future suites. It should be weakened or revised if any of the following persists:

- (F1) a clear gauge-candidate scenario ceases to preserve the structural transport signature,
- (F2) a dangerous compensated attack begins to preserve null-curvature and quotient-style Čencov stability simultaneously,
- (F3) hard structural violations cease to fail the structural signature,
- (F4) cross-model batteries show that the present structural signature is coil-specific rather than transport-specific,
- (F5) laboratory deployment contradicts the tested-class hierarchy observed here.

Proposition 5 (Tested-class falsifiability). *If any falsifier (F1)–(F5) occurs persistently on strengthened suites, the present transport-rigidity package must be weakened, restricted, or abandoned in its current form.*

Proof 15. Each falsifier negates one of the core tested-class claims: gauge persistence, compensated structural split, hard-violation refusal, or the transport-specific character of the structural signature. Persistent failure therefore undermines the theorem package itself.

11 Reproducibility and supplementary material

11.1 Companion script

The synthetic battery is accompanied by the Python companion

```
ctmt_transport_identifiability_battery.py.
```

Its role is operational:

1. generate the attacked suite,
2. compute the observable screen and structural indicators,
3. emit the scenario summary and aggregate table,
4. generate the dashboard figure and machine-readable summaries.

11.2 Expected artifacts

The companion is intended to generate at least:

- `ctmt_transport_identifiability_summary_final.csv`,
- `ctmt_transport_identifiability_invariant_table_final.csv`,
- `ctmt_transport_identifiability_dashboard_final.png`,
- `ctmt_transport_identifiability_run_summary_final.txt`,
- `ctmt_transport_identifiability_theorem_final.tex`.

12 Conclusion

The missing result was not another raw Fisher theorem. It was a tested-class transport-rigidity theorem.

The present attacked-coil battery now states that result cleanly. Baseline and current scaling remain in the same tested gauge-class. Radius scaling and compensated radius/current attack preserve a permissive observable screen and even survive Fisher monotonicity, winding, and self-reference contraction, yet remain non-gauge because null-curvature and quotient-style transport-class stability fail. Hard violations fail across the full battery.

This is exactly the kind of nontrivial synthetic result that makes the CTMT program more credible rather than less. It does not say that every invariant survives chaos. It says something more precise and more useful: under real transport-identifiability chaos, CTMT's presently strongest supported core is structural and gauge-relational, while Fisher-metric quantities are informative but not by themselves gauge-deciding.

A Supplementary Python Companion

The synthetic attacked-coil transport-identifiability battery is provided as the supplementary Python companion `ctmt_transport_identifiability_battery.py`. It reproduces the scenario summary, aggregate invariant table, dashboard figure, and machine-readable JSON/TXT outputs used in the present note.

B Reproducibility note

All numerical values reported in the main text are synthetic outputs of the attacked-coil battery.

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